

A Greener Jordan

South Jordan Tree Planting & Environmental Restoration Research

A short, practical research guide for choosing trees, planting them carefully, using as little extra water as possible, and protecting them after planting.

Core idea: Jordan does not need trees planted everywhere. It needs the right species in the right places, with the least possible long-term water demand.

1. Why water matters

Jordan faces severe water scarcity. The National Water Strategy 2023–2040 reports renewable freshwater availability of about 61 cubic metres per person per year, far below the commonly used 500 cubic metre absolute-scarcity threshold.

For this reason, A Greener Jordan should avoid projects that depend on permanent, high-volume irrigation. Newly planted trees still need establishment care, but the goal should be to reduce supplemental water as roots develop and to favor sites where rainfall and natural water movement can support long-term survival.

2. South Jordan: a careful, local approach

South Jordan is not one single climate. It includes hot, very dry lowlands and valleys as well as higher southern landscapes with different rainfall, soils and vegetation. A planting plan therefore has to be local rather than treating the whole south the same.

Priority areas to investigate include Ma'an, Tafila, Shobak, the Petra/Wadi Musa area, Aqaba and selected parts of the Desert Highway corridor. These should be treated as **candidate areas for study and permission**, not automatic planting sites.

For roads, the project should first identify safe public areas, ownership, existing vegetation, litter, soil, water availability and road-safety limits. Trees should never be planted where they create a traffic hazard or where volunteers cannot work safely.

3. Tree species to study first

Species	Why it is a candidate	South Jordan use
Carob (Ceratonia siliqua)	Mediterranean native; strong drought tolerance; useful for soil stabilization	Suitable sites with enough soil depth; establish with care, then monitor
Phoenician juniper (Juniperus phoenicea)	Native to Jordan; associated with southern highland forests	High southern sites with local Jordanian heritage; study natural distribution
Atlantic pistachio (Pistacia atlantica)	Native Jordanian species; drought-adapted and valuable for biodiversity	For biodiversity and restoration sites where soil and elevation fit
Evergreen oak (Quercus calliprinos)	Important Jordanian forest species and drought-resistant	More suitable on appropriate higher/rainier southern sites than carob
Aleppo pine (Pinus halepensis)	Native to Jordan; heat/drought tolerant and used in reforestation	Suitable reforestation sites after local soil, fire and water check

These are priority species to investigate, not a universal planting list. Species selection must follow local rainfall, elevation, soil, drainage, habitat and land-use conditions.

4. How to plant for low maintenance

Plant in the cool/wet season: Where appropriate, planting near the start of the rainy season can let rainfall help with establishment.

Make water work harder: Use small basins, contour micro-catchments or other suitable rainwater-harvesting methods. FAO has documented micro-catchment restoration approaches in Jordan's Badia.

Water deeply, not constantly: Newly planted trees need establishment water. The amount depends on tree size, soil, weather and rainfall. Do not use one fixed litres-per-tree number for every site.

Mulch: A suitable mulch layer can reduce evaporation and weeds. Keep mulch away from direct trunk contact.

Protect young trees: Grazing, trampling and damage can kill a tree even when water is adequate. Guards or fencing may be more valuable than extra irrigation.

Measure survival: Check trees at 3, 6, 12 and 24 months. The project should reward survival and healthy growth, not only planting-day numbers.

5. Watering guide

General tree-establishment guidance from university extension services commonly uses trunk size as one starting point; for example, about 1–1.5 US gallons (3.8–5.7 litres) per inch of trunk caliper per watering. This is a general guide, not a Jordan-specific lifetime water requirement.

A practical Jordan project should check the root-zone soil before watering, adapt to heat and wind, and reduce supplemental irrigation as the tree establishes. A Jordan-focused guide for Aleppo pine gives an example of 20–25 litres twice per week in the first year when planted after the rainy season, while noting no watering once established. This should be treated as a site-specific example rather than a rule for every species.

6. What success should mean

A Greener Jordan should track:

- Tree survival after 12 and 24 months
- Supplemental water used per surviving tree
- Area restored
- Native/local species share
- Volunteer hours
- Roadside and public-area waste removed
- Number of projects with documented landowner or authority approval

7. Environmental benefits

Well-designed planting can stabilize soil, reduce erosion, provide shade, support wildlife and improve the condition of public spaces. Vegetation can also change water movement, so tree planting should be planned at landscape scale rather than assuming that every tree automatically improves water availability.

The strongest project model is therefore **restoration + water conservation + community action**.

8. Main sources

- **Jordan National Water Strategy 2023–2040**

https://www.mwi.gov.jo/EBV4.0/Root_Storage/AR/EB_Ticker/National_Water_Strategy_2023-2040_Summary-English_-ver2.pdf

- **FAO — Responsible management of planted forests**

<https://www.fao.org/4/j9256e/j9256E06.htm>

- **FAO — Rainwater harvesting / Vallerani micro-catchments in Jordan**

<https://www.fao.org/family-farming/detail/en/c/1040697/>

- **Restoration Ecology (2026) — 23-year arboretum study in Jordan**

<https://onlinelibrary.wiley.com/doi/10.1111/rec.70178>

- **Jordan National Biodiversity Strategy and Action Plan**

<https://www.cbd.int/doc/world/jo/jo-nbsap-v2-en.pdf>

- **Penn State Extension — Meeting the watering needs of a newly planted tree**

<https://extension.psu.edu/meeting-the-watering-needs-of-a-newly-planted-tree>

- **University of Minnesota Extension — Watering newly planted trees and shrubs**

<https://extension.umn.edu/garden-and-home/yard-and-garden/gardening-in-minnesota/watering-newly-planted-trees-and-shrubs>

- **Center for the Study of the Built Environment — Aleppo pine in Jordan**

<https://www.csbe.org/aleppo-pine-pinus-halepensis>

Prepared for A Greener Jordan. This document is a planning and research guide, not permission to plant on any specific land. All field projects should follow local authority, landowner and safety requirements.